

AYMAN PATEL

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Professional Summary

Senior Frontend and Full-Stack Engineer with 6+ years building production platforms at Mastercard across **React, TypeScript, Redux, Spring, APIs, event-driven data pipelines, and databases**. Led a reusable microfrontend architecture supporting **10+ product teams**, improved UI load performance by **30%**, and raised WCAG compliance by **40%**. Owns delivery end to end – requirements, software architecture, UX, implementation, testing, releases, observability, and production incident resolution – while aligning product managers and engineering stakeholders across distributed teams.

Technical Skills

Frontend: React, TypeScript, JavaScript (ES6+), Redux Toolkit, RTK Query, Next.js, Vite, Webpack, Module Federation
Backend & APIs: Java, Spring, Spring Data JPA, Hibernate, REST APIs, Microservices, API Gateway, Event Sourcing, CQRS

Data Pipelines & Databases: PostgreSQL, PostgreSQL JSONB, Oracle, Redis, Kafka, RabbitMQ, Debezium, Change Data Capture

Testing & Quality: Jest, React Testing Library, Playwright, Cypress, JUnit, Mockito, Testcontainers, TDD, k6

Delivery & Observability: Jenkins, Gradle, Docker, Kubernetes, AWS, Grafana, OpenTelemetry, Splunk, Dynatrace, InfluxDB

Frontend Standards: WCAG, axe-core, semantic HTML, ARIA, focus management, responsive performance, Figma

Work Experience

Mastercard

Mar 2024 – Aug 2025

Senior Software Engineer

Vadodara, India

- **Led the architecture and rollout of a reusable microfrontend platform** using Webpack Module Federation, splitting a monolithic React frontend into independently deployable modules for **10+ product teams** and cutting cross-team release coordination from days to hours.
- Built the host shell with **React, TypeScript, Redux Toolkit, and RTK Query** for complex state management and data fetching; defined shared dependency boundaries, runtime module discovery, routing, and versioning contracts so teams could release independently without breaking the integrated user experience.
- **Owned full-stack design and delivery** across React frontends, Spring/JPA/Hibernate APIs, PostgreSQL JSONB, Redis caching, and Kafka event-sourced flows; translated product requirements into technical specifications and drove cross-country execution.
- Improved UI load performance by **30%** through Akamai CDN and bundle-delivery work; instrumented client-side traces and Core Web Vitals, then used Splunk and Dynatrace to triage and debug production problems, reducing MTTR from days to hours.
- Applied Resilience4j Circuit Breaker and Retry patterns across **5+ microservices**, reducing cascading failures and improving system availability by **15%**; used k6 and InfluxDB to identify performance regressions before release.
- Led accessibility engineering across the product suite, lifting WCAG compliance by **40%**; coached **10+ engineering teams**, drove knowledge sharing, contributed five inner-source fixes to the internal design system, and wrote guidance circulated to **200 engineers**.

Mastercard

Mar 2022 – Feb 2024

Software Engineer II

Vadodara, India

- Built greenfield full-stack applications with **React, TypeScript, Redux, and RTK Query** alongside **Spring, Oracle, and Hibernate** services; served as a core maintainer for frontend performance, accessibility, and code quality.
- Created unit and end-to-end suites with Jest, React Testing Library, Playwright, and Cypress, reaching **95% test coverage** and cutting CI execution from **8 hours to 3 hours** through parallelisation, selective retries, and caching.
- Took ownership when no UX resource was available, producing high-fidelity Figma prototypes in **2 days** to align product managers and stakeholders before implementation.

Mastercard

Jul 2019 – Feb 2022

Software Engineer I

Vadodara, India

- Designed REST APIs and automated enterprise workflows with Camunda, translating complex rules into BPMN, DMN, and CMMN models and delivering through Jenkins CI/CD.
- Developed and maintained Hardware Security Module integration for MDES tokenisation, securing card-data transmission across mobile payment flows; triaged penetration-test findings and delivered risk-prioritised remediation.

Nebius Academy

Mar 2026 – May 2026

AI Engineering Fellow

London, UK

- Built production-oriented services including a FastAPI repository summariser, a LangGraph research agent with dataflow-integrity checks, and a shared MCP tool layer that let multiple clients discover and call the same tools dynamically.
- Containerised distributed training pipelines on Kubernetes and debugged model, networking, and GPU-performance issues using version-controlled infrastructure and structured profiling.

Selected Projects

Web Accessibility Repair Platform

React, FastAPI, Playwright, axe-core

- Architected an end-to-end platform that collected aligned HTML, accessibility-tree, visual, and axe-core data for **704 webpages**; generated schema-constrained repairs through React and FastAPI, improving page-level F1 by **34.1 points**.

Go Microservices

Go, REST, gRPC, Envoy, Kubernetes

- Built a progressive service platform spanning validated REST APIs, Swagger/OpenAPI, gRPC, and an Envoy-based service mesh with Kubernetes and WASM filters.

Education

Queen Mary University of London

London, UK

MSc Advanced Computer Science

Sep 2025 – Sep 2026

Sardar Vallabhbhai Institute of Technology

Vadodara, India

Bachelor of Computer Science, GPA: 7.9/10

Jul 2015 – May 2019

Writing, Leadership & Recognition

Published **40+ technical articles** on frontend architecture, system design, observability, and distributed data infrastructure; work featured in the eBPF newsletter. Produced company-wide whitepapers on accessibility and microfrontend architecture, earning CTO recognition and adoption as internal engineering references. Winner of the **Most Secure Build Award** at the 2026 {Tech: Europe} Hackathon.